Decoding the Taste:

What Makes a *Vinho Verde* Red Wine High Quality?

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Group 8

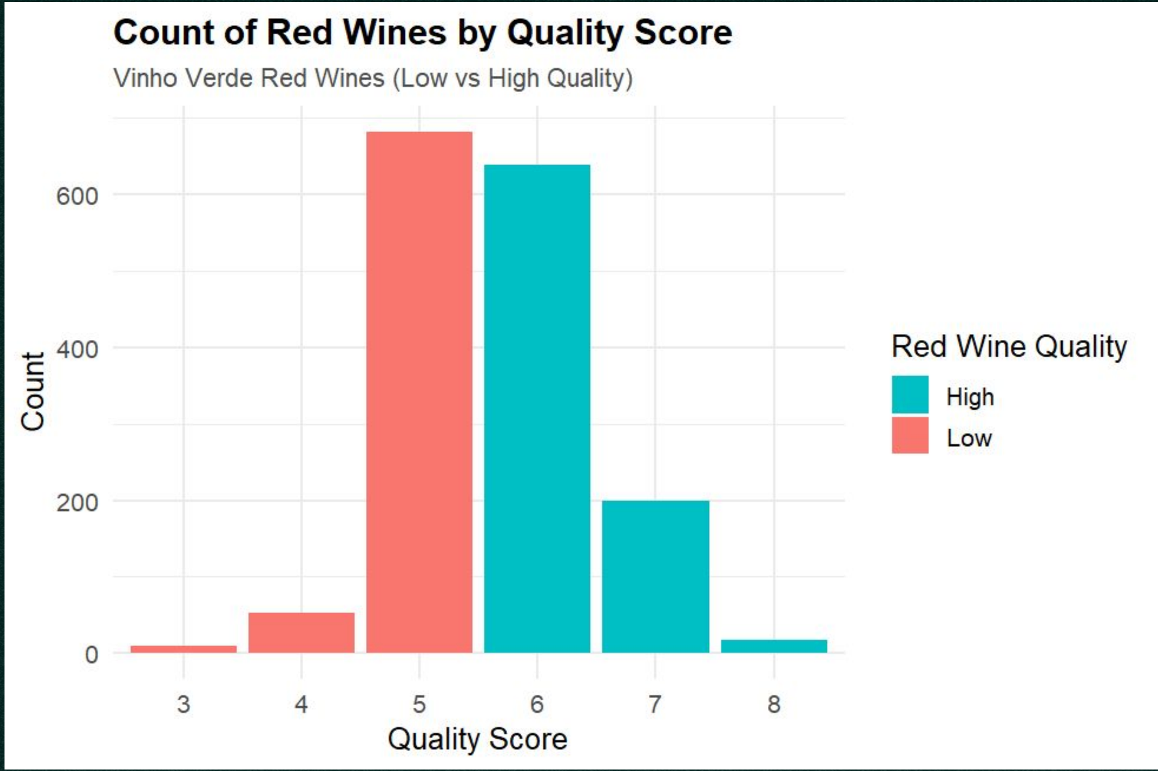
Introduction

Data Background

- Focus: Impact of alcohol and pH on red wine quality (Vinho Verde, Portugal).
- Dataset: 1,599 samples (2004–2007) with chemical properties + expert quality scores.
- Goal: Help producers optimize quality, retailers price accurately, and consumers choose wisely.

How does the % alcohol content and final pH level contribute to the quality score of *Vinho Verde* red wines?

Visual Representation

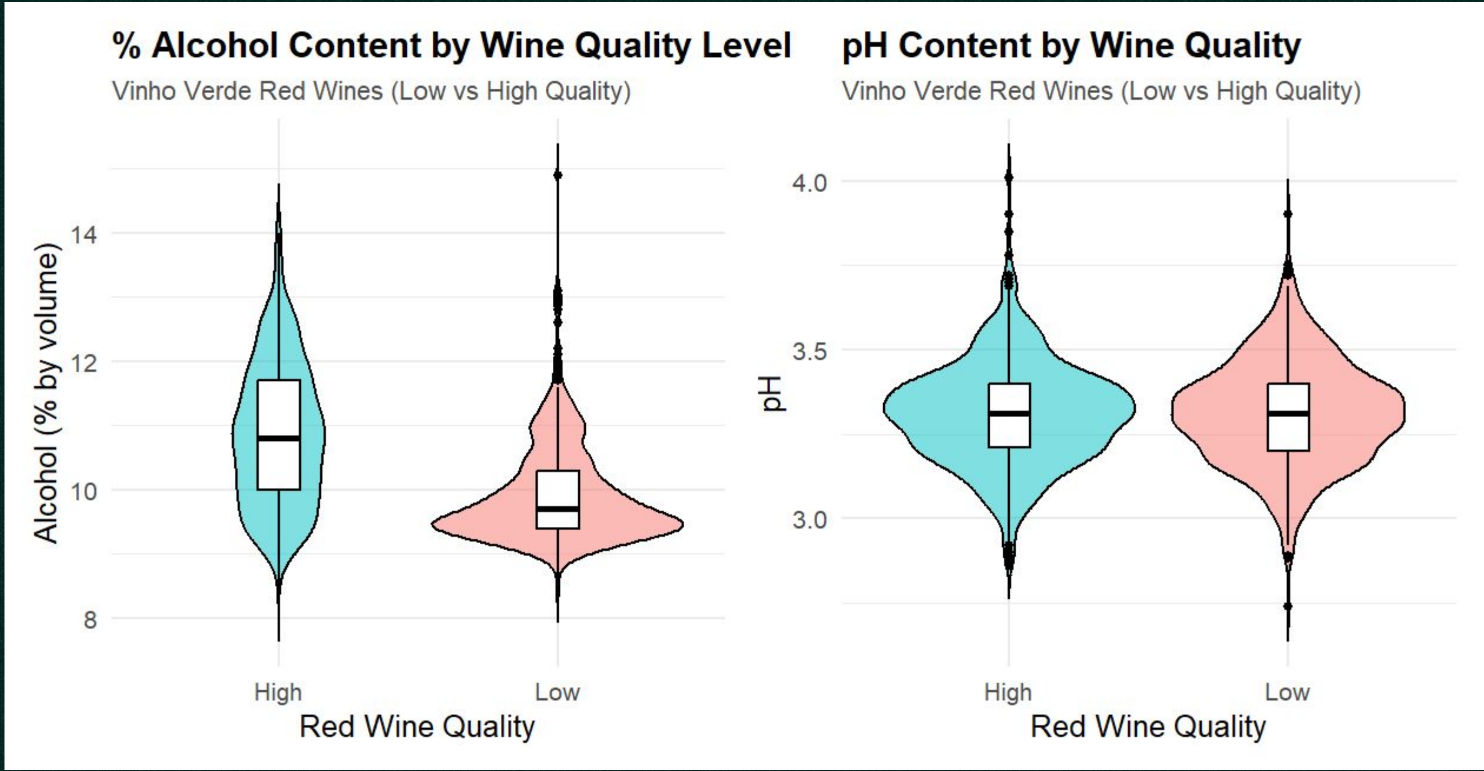


n = 1599

High quality = 855

Low quality = 744

Visual Representation



Question 1

Do high quality *Vinho Verde* red wines have a significantly higher alcohol content than low quality wines?

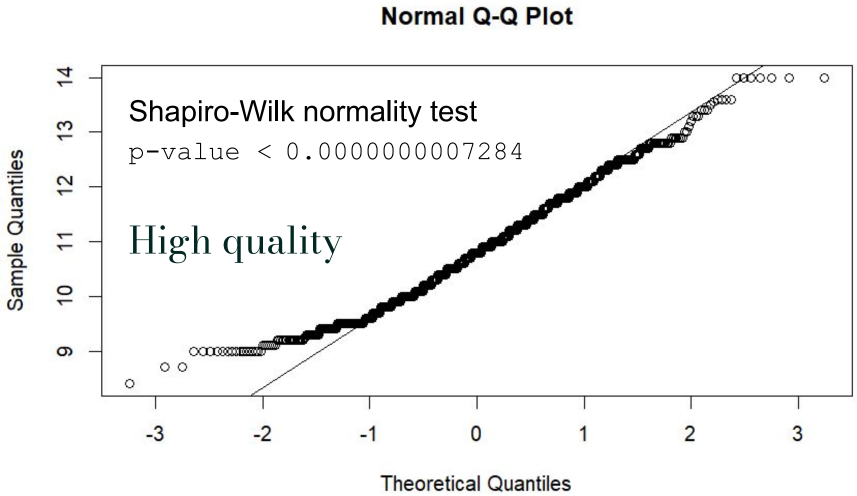
Statistical Hypotheses

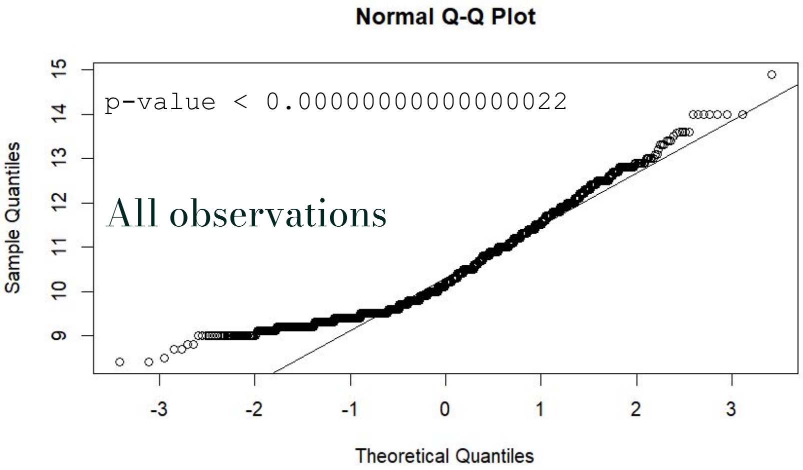
Null:

μ (High Quality) $\leq$ μ (Low Quality)

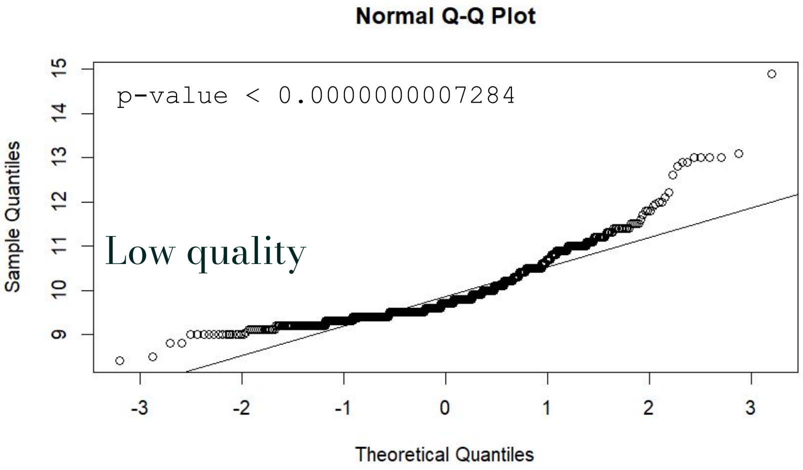
Alternate:

μ (High Quality) $>$ μ (Low Quality)

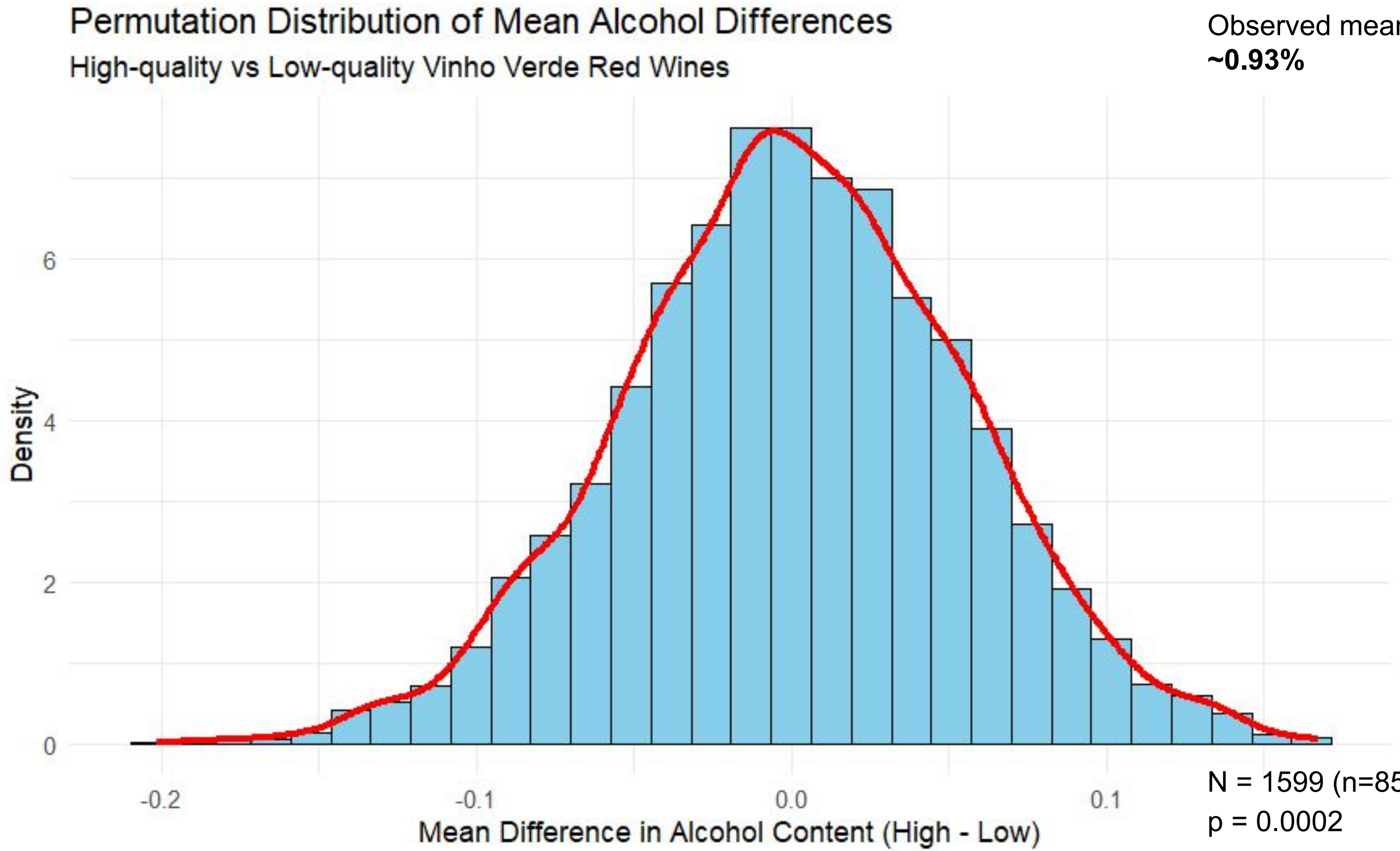




Normality Test



Permutation Test

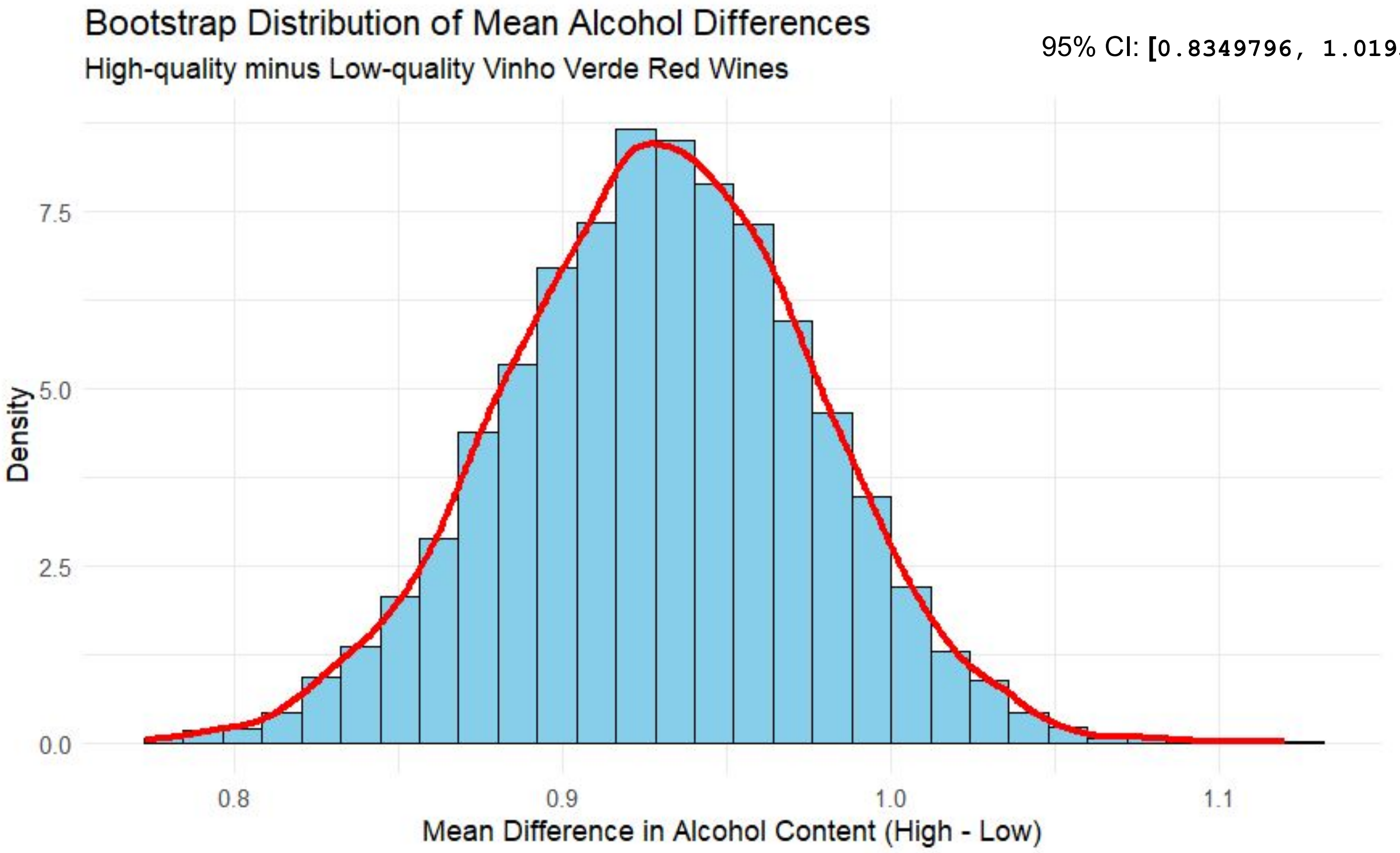


Observed mean diff.:
~0.93%

N = 1599 (n=855 high quality)
p = 0.0002

Bootstrap Confidence Intervals

95% CI: `[0.8349796, 1.0195235]`



Results Question 1

µ (High Quality) - µ (Low Quality) ~ 0.93%

95 % CI: [0.835, 1.02]

Permutation: $p = 0.0002$

High quality Vinho Verde red wines have **significantly more** alcohol % than low quality wines

Question 2

Is the pH level of the final *Vinho Verde* red wine significantly associated with its perceived quality?

Statistical Hypotheses

Null:
The quality of the red wine is independent from the pH level of the red wine

Alternative:
The quality of the red wine is not independent from the pH level of the red wine

pH Value Cutoffs

Chemical Cutoffs

High Acidity	< 3.2
Moderate Acidity	3.2 - 3.5
Low Acidity	> 3.5

Quartile Cutoffs

High Acidity	2.74 - 3.21
High Moderate Acidity	3.21 - 3.31
Low Moderate Acidity	3.31 - 3.40
Low Acidity	3.40 - 4.01

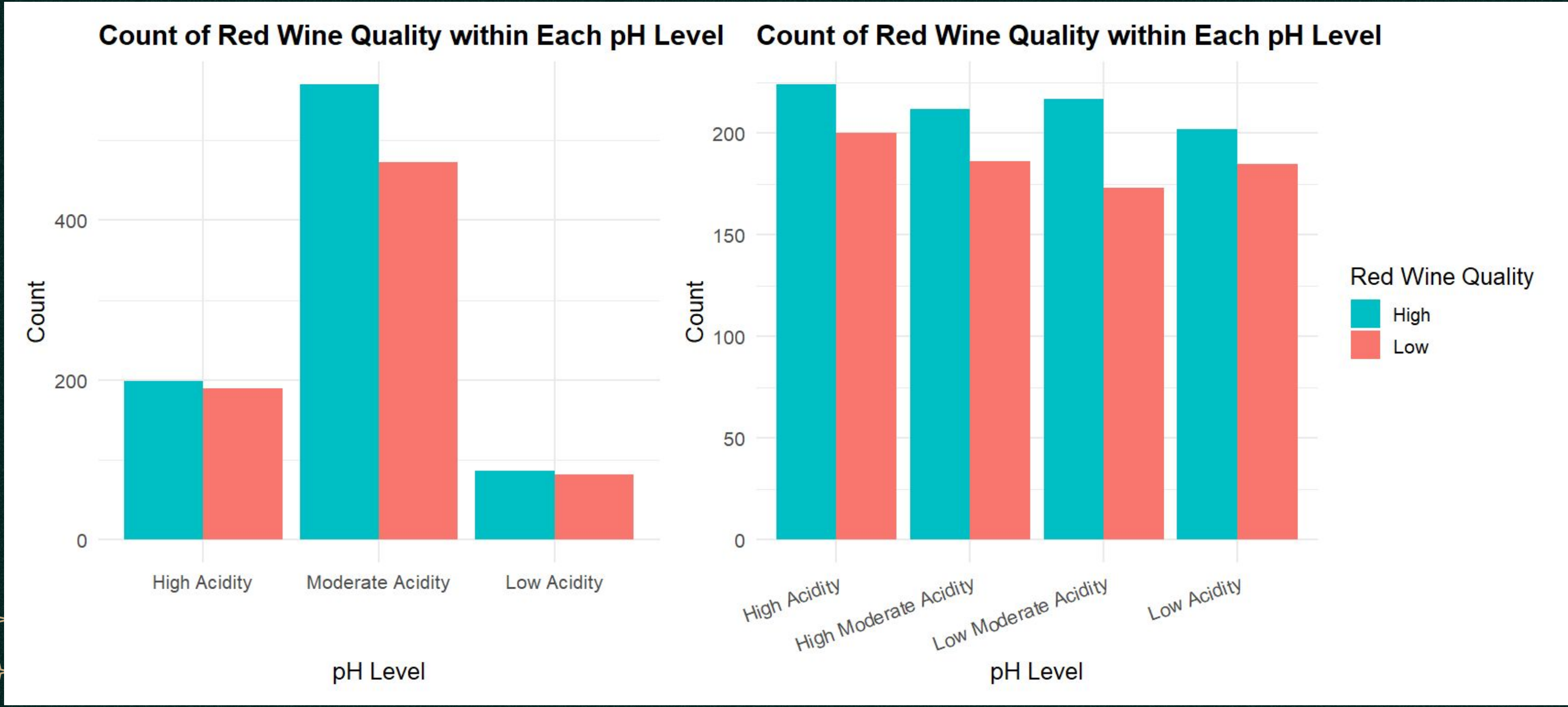
Atlas Scientific. (2023, July 7). The importance of pH in wine-making.
https://atlas-scientific.com/blog/the-importance-of-ph-in-wine-making/?srsltid=AfmBOop9MYjk5_0E7wn2nD5DSpdbH1rW2rsacn4zWHLYUgoJiSE7\-lx4

Cross Tables

Quality/Acidity	High	Moderate	Low
Low	189	473	82
High	199	570	86

Quality/Acidity	High	High Moderate	Low Moderate	Low
Low	200	186	173	185
High	224	212	217	202

Cross Tables



Chi-squared Tests

```
        Pearson's Chi-squared test

data:  x
X-squared = 1.6767, df = 2, p-value = 0.4324

 189.00   473.00    82.00
(180.53) (485.30) ( 78.17)
 [0.40]   [0.31]   [0.19]
< 0.63>  <-0.56>  < 0.43>

 199.00   570.00    86.00
(207.47) (557.70) ( 89.83)
 [0.35]   [0.27]   [0.16]
<-0.59>  < 0.52>  <-0.40>
```

Chemical Cutoffs

Quartile Cutoffs

```
        Pearson's Chi-squared test

data:  x
X-squared = 1.0676, df = 3, p-value = 0.7849

 200.00    186.00    173.00    185.00
(197.28)  (185.19)  (181.46)  (180.07)
[0.0374]  [0.0036]  [0.3947]  [0.1351]
< 0.193>  < 0.060>  <-0.628>  < 0.368>

 224.00    212.00    217.00    202.00
(226.72)  (212.81)  (208.54)  (206.93)
[0.0326]  [0.0031]  [0.3435]  [0.1176]
<-0.180>  <-0.056>  < 0.586>  <-0.343>
```

Results Question 2

The **p-values** from both Chi-squared test of independence (0.4324 and 0.7849) are **larger** than our set alpha value of **0.05**.

We **fail to reject** the null hypothesis

The **quality** of the *Vinho Verde* red wines **does not significantly depend on** its final **pH.**

Conclusion

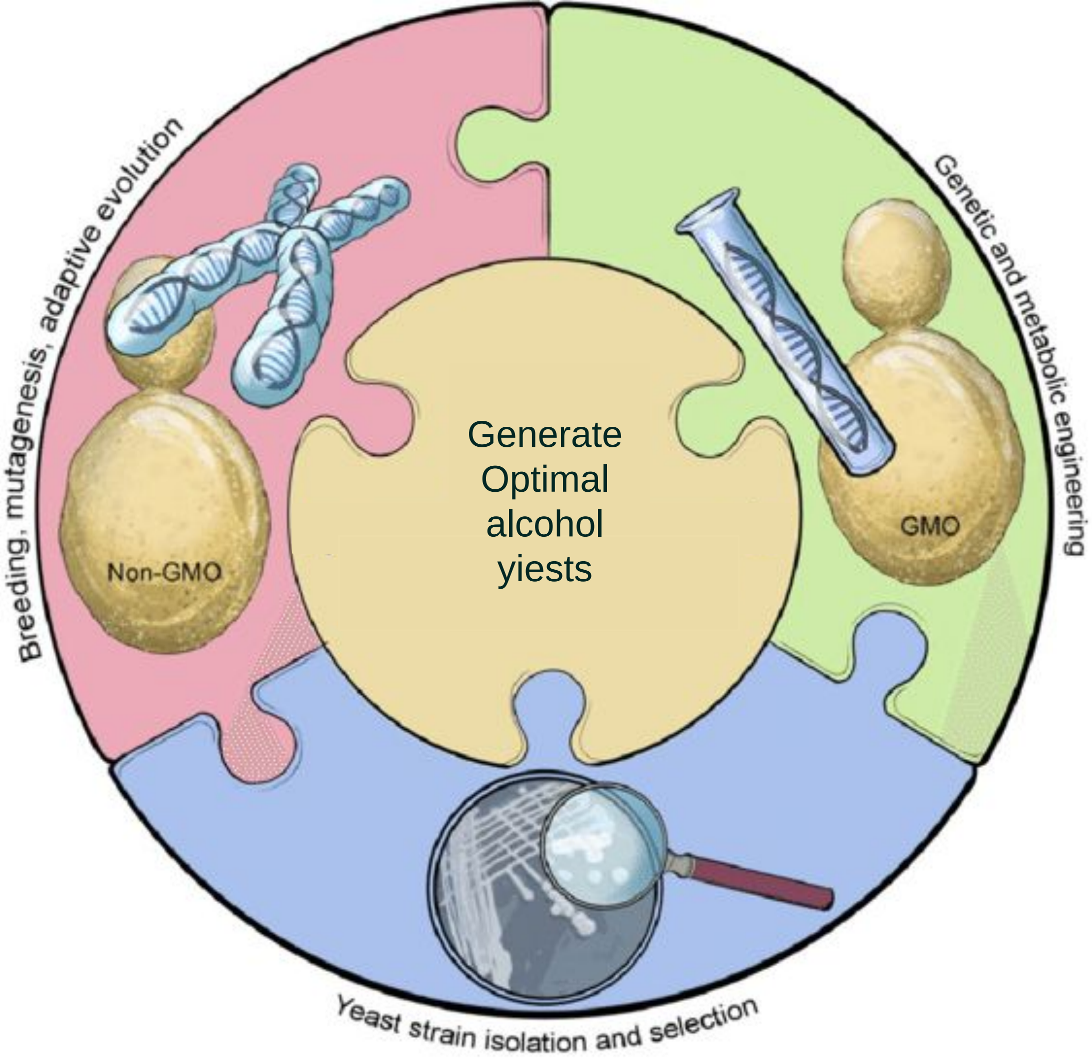
High quality *Vinho Verde* red wine has ~1% more alcohol content compared to low quality red wines while the final pH level does not have a great effect on red wine quality

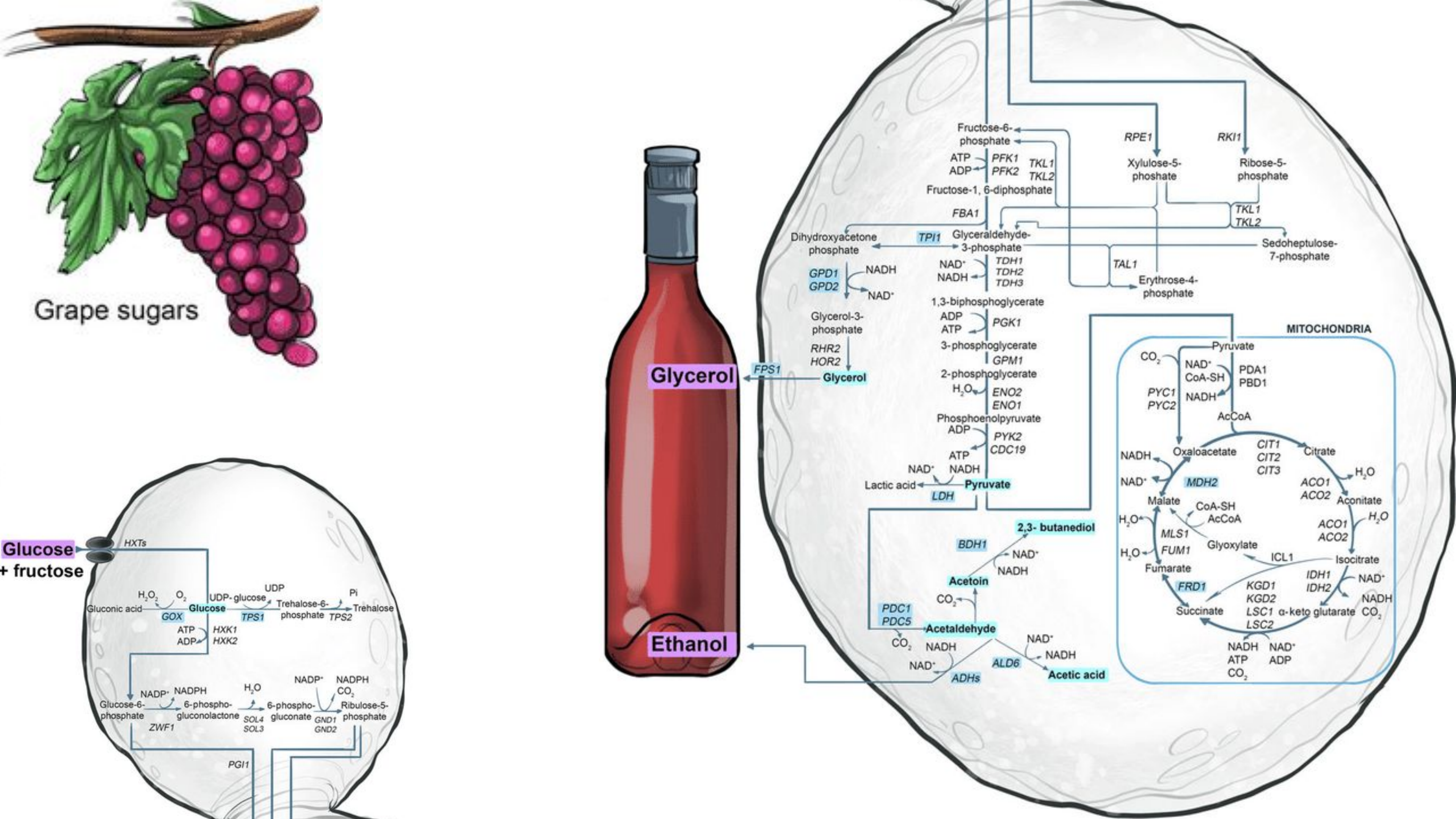
Limitations

- Data limited to red Vinho Verde wines (Portugal, 2004–2007)
- Missing info: grape type, vineyard condition, fermentation methods
- Possible subjective bias (tasting panel) in wine quality scores
- Simplistic methods (Chi-square, permutation tests)
- Only one dataset → limited verification

Recommendations & Future Directions

1. Include more variables (e.g., volatile acidity, sulphates, SO_2)
2. Use advanced methods (MLR, PCA, machine learning)
3. Expand to other regions & newer vintages
4. Conduct experimental studies (control pH, alcohol, fermentation)
5. Build predictive models for quality & consumer preferences


Breeding, mutagenesis, adaptive evolution
Genetic and metabolic engineering
Generate
Optimal
alcohol
yiests
Non-GMO
GMO
Yeast strain isolation and selection



Grape sugars
Glucose
+ fructose
HXTs
H_2O_2
O_2
Gluconic acid
GOX
Glucose
UDP- glucose
UDP
TPS1
Trehalose-6-
phosphate
Pi
TPS2
Trehalose
ATP
ADP
HXK1
HXK2
Glucose-6-
phosphate
$NADP^+$
NADPH
ZWF1
6-phospho-
gluconolactone
H_2O
SOL4
SOL3
6-phospho-
gluconate
$NADP^+$
NADPH
CO_2
GND1
GND2
Ribulose-5-
phosphate
PGI1
Glycerol
FPS1
Ethanol
Fructose-6-
phosphate
ATP
ADP
PFK1
PFK2
TKL1
TKL2
Fructose-1, 6-diphosphate
FBA1
RPE1
Xylulose-5-
phoshate
RKI1
Ribose-5-
phosphate
TKL1
TKL2
Dihydroxyacetone
phosphate
TPI1
Glyceraldehyde-
3-phosphate
Sedoheptulose-
7-phosphate
TAL1
Erythrose-4-
phosphate
GPD1
GPD2
NADH
NAD^+
Glycerol-3-
phosphate
RHR2
HOR2
Glycerol
NAD^+
NADH
TDH1
TDH2
TDH3
1,3-biphosphoglycerate
ADP
ATP
PGK1
3-phosphoglycerate
GPM1
2-phosphoglycerate
H_2O
ENO2
ENO1
Phosphoenolpyruvate
ADP
PYK2
CDC19
ATP
NAD^+
NADH
Lactic acid
Pyruvate
LDH
2,3- butanediol
BDH1
NAD^+
NADH
Acetoin
CO_2
PDC1
PDC5
Acetaldehyde
CO_2
NADH
NAD^+
ADHs
ALD6
NAD^+
NADH
Acetic acid
MITOCHONDRIA
Pyruvate
CO_2
NAD^+
CoA-SH
NADH
PDA1
PBD1
PYC1
PYC2
AcCoA
CIT1
CIT2
CIT3
Oxaloacetate
Citrate
NADH
NAD^+
MDH2
H_2O
ACO1
ACO2
Aconitate
Malate
CoA-SH
AcCoA
H_2O
ACO1
ACO2
H_2O
MLS1
FUM1
Glyoxylate
H_2O
ICL1
Isocitrate
Fumarate
IDH1
IDH2
NAD^+
NADH
CO_2
FRD1
KGD1
KGD2
LSC1
LSC2
Succinate
α-keto glutarate
NADH
ATP
CO_2
NAD^+
ADP

Thanks!

References

Dataset available at: Cortez, P. (2017). Wine quality – red wine dataset [Data set]. Kaggle. https://www.kaggle.com/datasets/uciml/red-wine-quality-cortez-et-al-2009

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